

"Chemistry of Transition Elements, Lanthanides and Actinides"

d-block elements:

The elements of periodic table belonging to group 3 to 12 are known as d-block elements. In other words, the elements in which last electron enters in d-sub shell or d-orbital are called d-block elements.

→ d-block elements lie in between s- and p-block elements in the long form of periodic table.

→ d-block elements show intermediate properties between s- and p-block elements.

→ Number of d-block elements are 41 with Thorium (90).

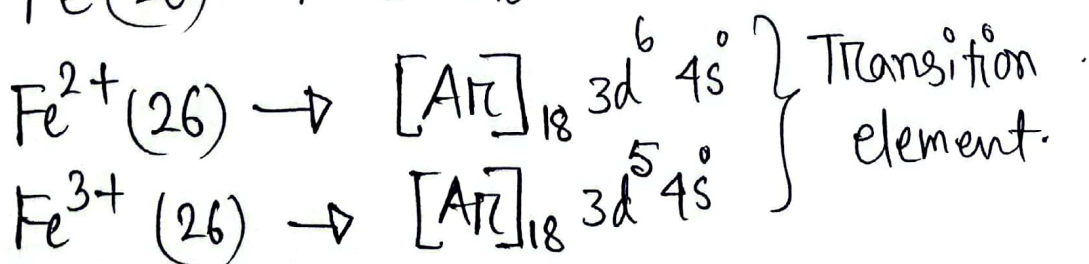
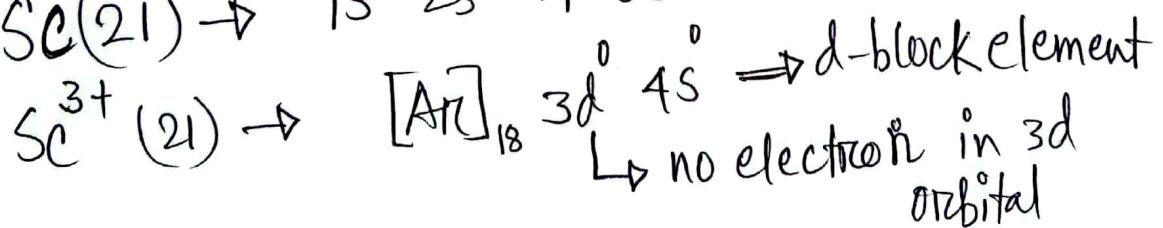
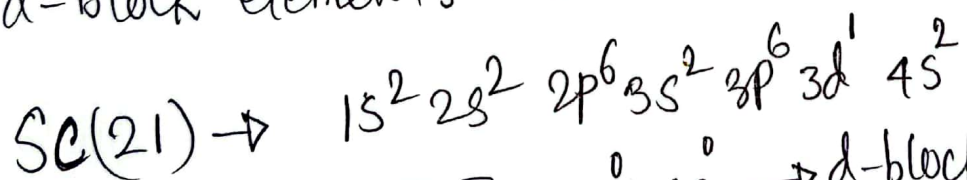
⇒ General electronic configuration of d-block elements: $(n-1)d^{1-10} ns^{1-2}$.

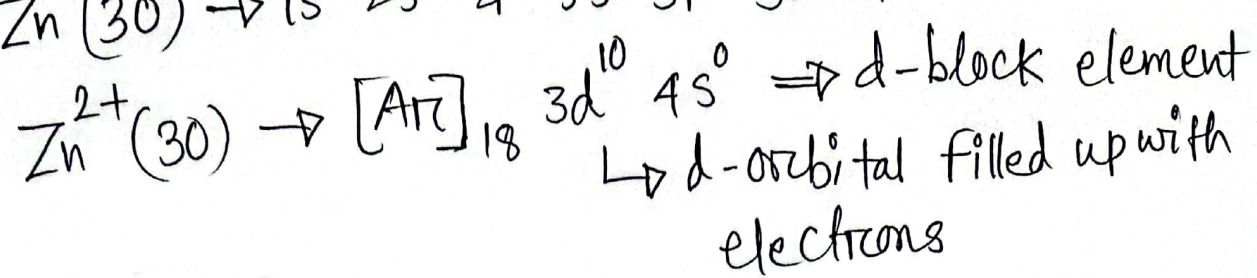
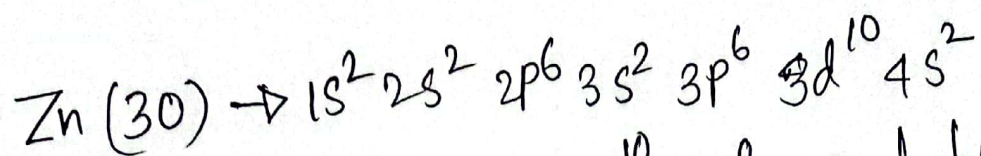
⇒ Transition elements :

d-block elements which form at least one ion with a partially filled sub-shell of d electrons i.e., d^1-d^9 , are called transition elements or transition metals.

✗ All d-block elements are not transition elements but all transition elements are d-block elements.

Examples of some transition elements and d-block elements with their electronic Confⁿ.





Ex. General Properties of Transition Elements:

1. Show variable valency
2. Form coloured compound
3. Form complex compound
4. Used as catalysts.
5. Show paramagnetic properties.

⇒ 1. Different Oxidation States: Variable Valency

The term Oxidation state is preferred to valency. Most of the transition metals show several oxidation states or variable valency in their compounds.

→ Oxidation states, electronic confⁿ in valence shell and some compounds of 3d-block elements are shown below:

$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s^1$
Ti	V	Cr	Mn	Fe	Co	Ni	Cu
(+2)	+2	+2	+2	+2	+2	+2	+1
+3	+3	+3	(+3)	+3	+3	+3	+2
+4	+4	(+4)	+4	(+4)	(+4)		
	+5	(+5)					
		+6	(+6)	(+6)			
			+7				

(Note: Oxidation states shown in brackets are exhibited in a few compounds.)

Different compounds of Cr: $CrCl_2$ $CrCl_3$ K_2CrO_4 $K_2Cr_2O_7$

Oxidation states of Cr: +2 +3 +6 +6

Different compounds of Fe: FeO Fe_2O_3

Oxidation states of Fe: +2 +3.

2. Formation of Coloured Compounds:

Normally the compounds of s- and p-block elements are colourless while those of transition metals in the solid or in solution state are usually coloured.

The colours of hydrated cations of transition metals of the first transition series (3d-series)

Sc^{3+} ($3d^0$, $n=0$, d-block element, colourless.)

Ti^{4+} ($3d^0$, $n=0$, colourless)

Ti^{3+} ($3d^1$, $n=1$, purple)

V^{3+} ($3d^2$, $n=2$, green), V^{2+} ($3d^3$, $n=3$, violet)

Cr^{3+} ($3d^3$, $n=3$, green), Cr^{2+} ($3d^4$, $n=4$, blue)

Mn^{3+} ($3d^4$, $n=4$, violet), Mn^{2+} ($3d^5$, $n=5$, faint blue or green).

Cu^{2+} ($3d^9$, $n=1$, blue), Cu^{+} ($3d^{10}$, $n=0$, colourless)

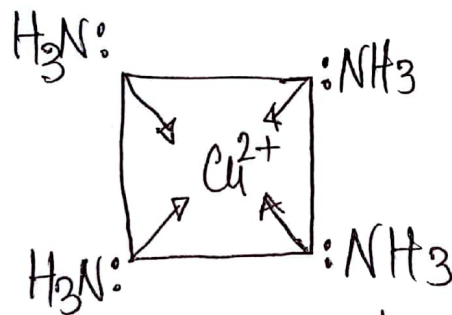
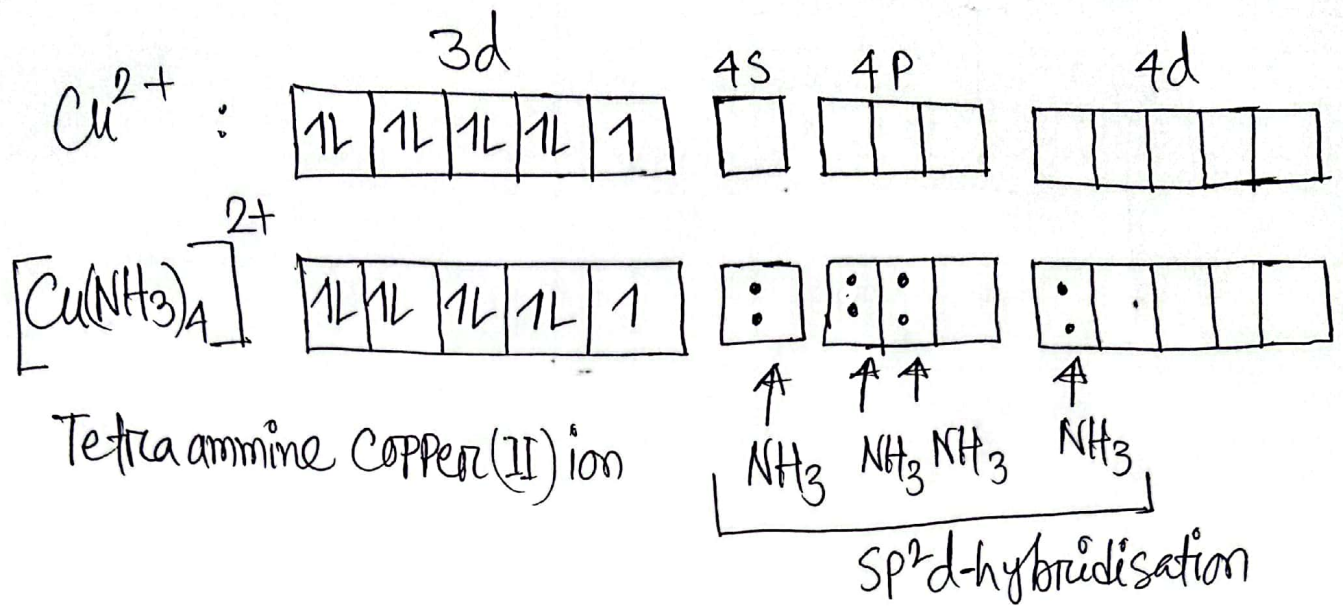
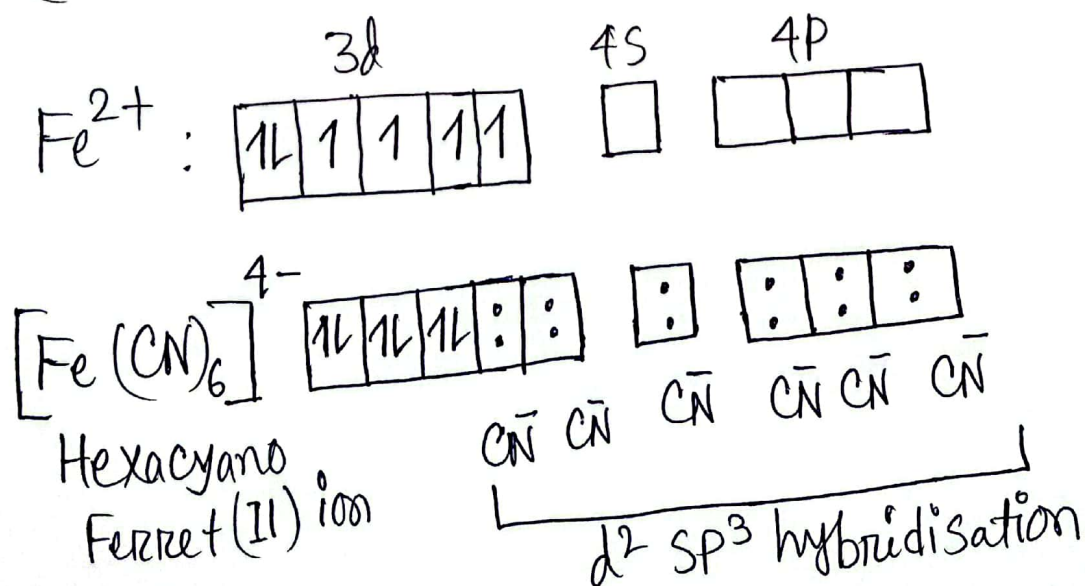
Here, n denotes the number of unpaired electrons in the respective cations.

3. Formation of Complex Compounds:

The cations of d-block elements have a strong tendency to form complexes with certain molecules (e.g. CO , NO , NH_3 etc.) or ions (e.g. F^- , Cl^- , CN^- etc.)

- The molecules or ions with which the cations attach themselves to form the complexes are called ligands.
- The molecules or ions acting as ligands have one or more 'lp' of electrons.
- The cations are relatively very small in size and hence have high positive charge density which makes it easy to accept the 'lp' of electrons from ligands.
- The cations have vacant $(n-1)d$ orbitals which are of appropriate energy to accept 'lp' of electrons from the ligands for bonding with them.

(A)

Example: Formation $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex:
 $[\text{Cu}(\text{NH}_3)_4]^{2+} \rightarrow \text{Square Planar.}$
(B.) Formation of $[\text{Fe}(\text{CN})_6]^{4-}$ complex:

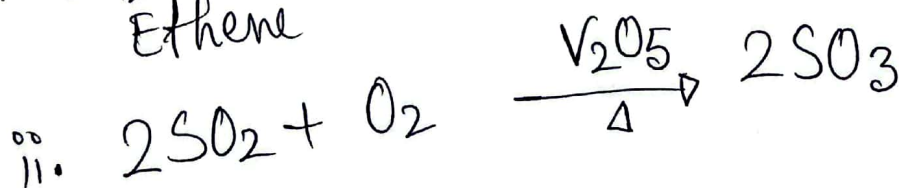
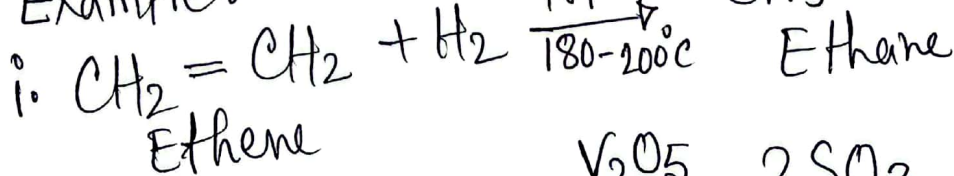


Most of the transition metals and their compounds are used as catalysts. Transition metals show a variety of oxidation states and thereby can form intermediate products with various reactants.

Example:

1. $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \xrightarrow[180-200^\circ\text{C}]{\text{Ni}^\circ} \text{CH}_3 - \text{CH}_3$
Ethene Ethane

V205 2 SM2



5. Paramagnetic Properties:

Transition metals show paramagnetic properties. The substance which shows paramagnetic properties is known as paramagnetic substance and the attraction of paramagnetic substance to the magnetic field is called paramagnetism.

Transition metals have unpaired electrons show paramagnetic properties.

e.g., Ti^{3+} , V^{3+} , Co^{3+} , Mn^{2+} , Fe^{2+} , Fe^{3+} , Ni^{2+} , Cu^{2+} etc. show paramagnetic properties.

Lanthanides & Actinides

→ 4f-Series (First Inner Transition Series):

The differentiating electron enters in 4f orbitals. The elements belonging to this series are also known as Lanthanides or Lanthanones.

→ 5f-Series (Second Inner Transition Series):

The differentiating electron enters in 5f orbitals. The elements belonging to this series are called Actinides.

⇒ Inner Transition Elements:

The elements in which the filling of atomic orbitals by electrons take place in f-subshells, two levels inside the outer subshell, are known as inner transition elements.

In other words, the f-block elements which form at least one stable ion with partially filled f-orbital with electrons. i.e., $f^1 - f^{13}$, are called inner transition elements.

⇒ Lanthanides:

The fifteen elements from Lanthanum (At no. 57) to Lutetium (At no. 71) lies in group 3 and Period 6 are known as Lanthanides or rare earth elements.

Electronic Configuration:

The general electronic configuration of these elements is $[Xe]_{54} 4f^{0-14} 5d^{0-1} 6s^2$

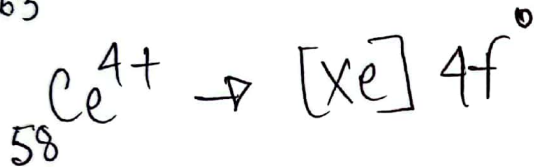
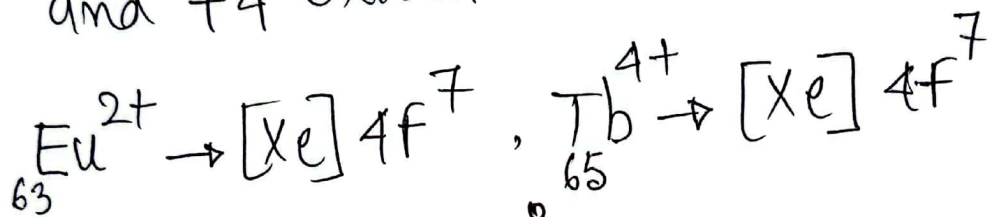
→ La (57): $[Xe] 4f^0 5d^1 6s^2$

→ Lu (71): $[Xe] 4f^{14} 5d^1 6s^2$.

⇒ General Properties of Lanthanides:

1. Oxidation State:

The common oxidation state of lanthanides is +3 but some elements exhibit +2 and +4 oxidation states.



2. Magnetic Properties:

All lanthanide ions with the exception of La^{3+} , Lu^{3+} and Ce^{4+} are paramagnetic in nature.

3. Lanthanide Contraction:

There is a steady decrease in the radii as the atomic number of the lanthanide elements increase. The contraction in size is quite regular and known as lanthanide contraction.

4. Colour:

The species containing unpaired electrons are coloured and so is the case with lanthanides ions. The f-f transitions are possible due to absorption of light from the visible region.

5. Melting and Boiling Point:

Lanthanides have high melting and boiling points.

6. Density:

Lanthanides have densities varying from. But there is no definite trend for these values.

7. Electronegativity:

Electronegativity values of lanthanides are almost same as that of s-block elements. Lanthanides form ionic compounds.

8. Ionization energies:

The ionization energy values of lanthanides are not very high due to their large size.

9. Complex Compound:

Due to having large ionic size they have little tendency to form complexes.

10. Reactivity:

Due to their low values of ionization energies the lanthanides are very reactive.

⇒ Actinides: The fifteen elements from actinium (At. no. 89) to lawrencium (At. no 103) are known as actinides and constitute the 5f series. From neptunium to onwards the elements are man made (artificially prepared) and also known as transuranium elements.

⇒ Electronic Configuration:

The differentiating electron enters the 5f atomic orbital. Their general electronic confⁿ is $[Rn] 5f^{0-14} 6d^{0-2} 7s^2$.

1. ⇒ General Properties of Actinides:

→ Oxidation State:

The common oxidation state is +3 but other oxidation states also exhibited by actinides, the maximum being +7.

2. Actinide Contraction:

It is similar to lanthanide contraction due to poor shielding of electrons.

3. Melting and Boiling points:

They have high values for melting and boiling points but there is no trend.

4. Density:

The value of density vary from 7.0 g cm^{-3} to 19.84 g cm^{-3} but there is also no regular trend.

5. Reactivity:

Actinides are very reactive in nature and combine with oxygen and halogens like lanthanides.

6. Coloured ions:

Actinide ions are coloured due to the presence of unpaired electrons and transitions.

7. Complex Formation:

They have higher tendency to form complex compounds.

